Kaustubh Hemant Sonawane

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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

Machine Learning Engineer with **3 years** of industry experience building and operating **LLM and agent-based systems** (**RAG** and **tool-calling agents**) in a regulated financial services setting, where behaviour was guardrailed through prompts versioned in **Git** and every release was reviewed by **Compliance**. Designed a **calibrated scoring and ranking algorithm** and its **data-preprocessing pipeline** in an **AstraZeneca** dissertation for an **MSc in Data Science at the University of Bristol** (awaiting result), and completed a **multi-architecture deep learning study** evaluated across five seeds and against pre-trained VLMs. Seeking a **Machine Learning Engineer** role owning **evaluation of LLM and agent systems**, weighing quality against cost and latency.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

LLM & Agent Systems in Production

- Designed and built production **LLM and RAG** applications end to end (**ReactJS** frontend, **Python FastAPI** microservices, retrieval with **LlamaIndex** and **Azure OpenAI GPT-3.5 & Llama-3**) for Compliance, Client Service and Investment, with a separate **pgvector** database and **Azure OpenAI** embeddings per department.
- Built **agentic tool-calling** where AI agents call **Python functions** as tools to run SQL-backed fund comparisons and generate PDF reports on request, on top of a **PostgreSQL** platform queried with **Python Pandas**.
- Took applications from prototype to real users and iterated on what broke; guardrailed behaviour through **system prompts** (no fine-tuning), versioned every **prompt change in Git** and had **Compliance** review each release; weighed cost against quality by **self-hosting Llama-3 on a GPU EC2 instance** for the largest department and using stronger **Azure OpenAI** outputs for decision makers.

MLOps & Deployment

- Designed the infrastructure architecture for internal analytics platforms and GenAI applications on **AWS** (**EC2 including a GPU instance, S3, ECR and Lambda**), packaged the **FastAPI** microservices as **Docker** containers and automated deployment of the separate frontend and backend repositories (dev and prod) through a one-click **Jenkins and GitLab CI/CD** pipeline.
- Implemented **RBAC** with **LDAP** authentication, restricted access to company private IPs with **AWS Security Groups** and deployed an **Nginx** reverse proxy, one of the fixes that took the applications from repeatedly failing to passing the security team's **VAPT**.

Data Pipelines, Analytics & Stakeholder Delivery

- Designed a **Normalized SQL Schema** (**PostgreSQL**) and wrote **Python** ingestion scripts for the **Morningstar API** covering every fund in the Indian mutual fund market (2023–2025), refreshed fortnightly and monthly and integrated with legacy analyst data; developed **Optimized SQL** queries, **SQL Functions** and **Procedures**.
- Built a **Python** extraction pipeline that parses over **300 daily emails** from credit rating agencies into structured data, fuzzy-matches them to tracked companies and sends analysts a daily summary, saving **60 analyst hours a**

month; scheduled on **AWS SageMaker** with **EventBridge** and **Lambda**, and replaced **AWS Textract** with **Azure OpenAI** when a rating agency began sending emails as images.

- Analyzed customer behaviour using **Python Pandas** and **Market Basket Analysis (Apriori Algorithm)** to surface buying patterns for the marketing team; migrated financial analysts from Excel to **Power BI** dashboards, improving efficiency by **10 hours a week**.
- Worked directly with stakeholders in **Client Service, Compliance and Investment** and alongside engineers and interns to turn requirements into working tools; **de facto team lead for up to 4 interns**; **Best Team Project** twice (2024 rating-email automation, 2025 department RAG chatbot).

PROJECTS

- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built. My focus: the **scoring and ranking algorithm** that ranks cancer cell lines by target inclusion and exclusion genes using per-gene scores across six evidence layers, calibrated across lineages and grouped into confidence tiers; the **data preprocessing pipeline** that builds 14 clean, joined tables from 17 raw inputs with integrity checks; and the **RAG-based AI assistant**. Ranking validated with feedback from AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **University of Bristol MSc Applied Data Science Group Project:** VisionCaptioner (Team project. **Multi-architecture deep learning study** comparing CNN, ResNet and ViT encoders trained from scratch with symbolic, SBERT-retrieval and LSTM/attention decoders for generating image descriptions on three synthetic datasets, **evaluated across five seeds** with SBERT cosine similarity, BLEU and ROUGE-L and benchmarked against BLIP/SigLIP and commercial VLMs, noting that the protocols differ so the comparison is not like-for-like.) GitHub: https://github.com/Arishtanemi4/ata-team30.git
- **Wingman:** AI dating-coach web app with **108 LLM personas** across 12 archetypes and a vision model that scores uploaded photos (ReactJS, Python FastAPI, NVIDIA NIM). LLM output treated as untrusted: **validated with Pydantic, repaired and cached by input hash**; cost and rate limits shaped the design (12 LLM calls instead of 108, results streamed to the UI). GitHub: https://github.com/Arishtanemi4/wingman.git
- **B.Tech Final Year Project:** Udacity self-driving car simulator. (A self-driving car trained using a **Convolutional Neural Network (CNN)** model.)

ACHIEVEMENTS

- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle Credit card fraud detection problem.)

SKILLS

- **Tools:** Jupyter, Anaconda, AWS (EC2 incl. GPU, S3, ECR, Lambda, SageMaker, EventBridge), Azure OpenAI, LlamaIndex, Docker, Jenkins, GitLab, Git, Nginx, Linux, LDAP, Power BI, VS Code, Claude Code.
- **Languages:** Python (pandas, scikit-learn, FastAPI, Pydantic, uvicorn), SQL, R (dplyr, tidyverse, ggplot2), NodeJS (express, ReactJS), Shell
- **Databases:** PostgreSQL (pgvector, SQL functions and procedures), SQLite